

How Backtest Overfitting in Finance Leads to False Discoveries

A Critical Analysis of Statistical Mirage in Investment Strategy Design

Based on: Bailey, D.H. & López de Prado, M. (2021). *Significance*, 18(6), 22–25.

DATA 498 – Senior Capstone

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Background

Investment funds seek to beat the market without taking on more risk. To find “optimal” strategies, analysts feed historical data into algorithms testing **millions of parameter combinations** – called **backtesting**.

Why Finance Is Especially Vulnerable

- **Low base rate:** Truly profitable strategies are rare due to intense market competition
- **Short-lived truth:** Real findings fade quickly as markets adapt
- **No controlled experiments:** Unlike medicine, strategies cannot be randomized – false claims are nearly impossible to debunk

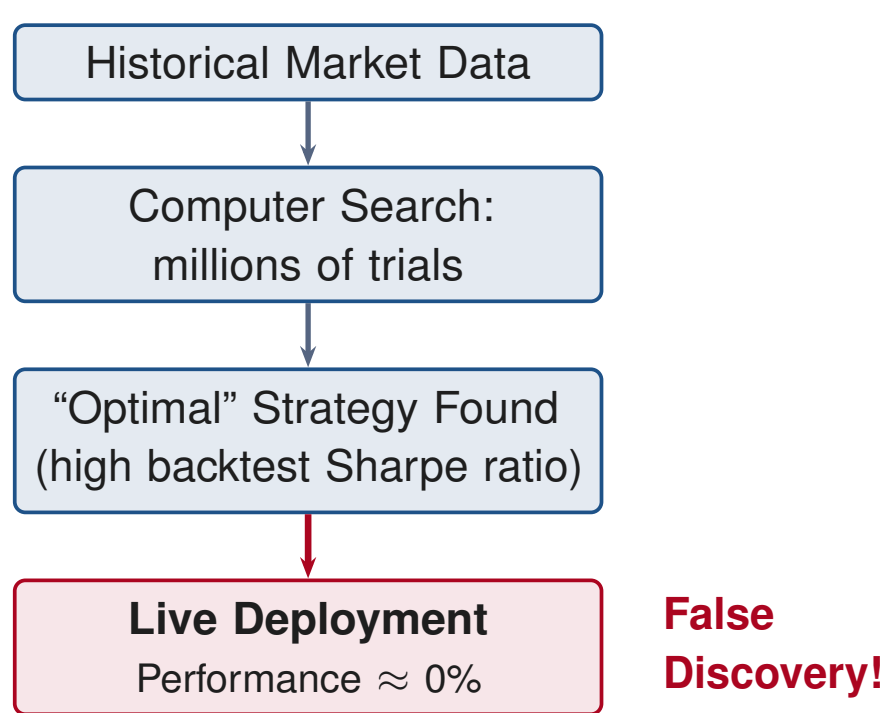


Figure 1: The backtesting trap.

Backtest Overfitting: Using historical market data to design a strategy where *too many variations are tried relative to the amount of data available*. A form of selection bias under multiple testing. (Bailey & de Prado, 2021)

Research Question

“How widespread is backtest overfitting in finance, and what statistical tools can detect and correct for this form of false discovery?”

Sub-questions addressed:

- How does computer-based strategy search amplify overfitting risk?
- Do real ETF and fund data confirm overfitting?
- Can a corrected Sharpe ratio separate true from false strategies?

Bailey & de Prado analyzed **68 professional market forecasters** over 17 years and found average accuracy of only **48%** – statistically indistinguishable from chance.

References

[1] Bailey, D.H. & López de Prado, M. (2021). How backtest overfitting leads to false discoveries. *Significance*, 18(6), 22–25.

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[3] Harvey, C.R. (2017). Presidential address. Duke I&E Research Paper No. 2017-05.

[4] Hou, K., Xue, C. & Zhang, L. (2020). Replicating anomalies. *Review of Financial Studies*, 33, 2019–2133.

[5] McLean, R.D. & Pontiff, J. (2015). *Journal of Finance*, 71(1), 5–32.

[6] Sharpe, W.F. (1994). The Sharpe ratio. *Journal of Portfolio Management*, 21(1), 49–58. etc.

Methods & Mechanism

The Monthly Cycle Example

Even a *simple* strategy has an enormous parameter space:

Backtest time period	Countless
Portfolio composition	Countless
Stock weights	Countless
Buy day of month (D1)	29 choices
Sell day (D2 > D1)	435 pairs

Even just timing parameters give **435 independent choices**. Adding portfolio composition pushes trials into the **millions**. Every extra trial raises the probability of a spurious result.

The False Strategy Theorem

Given K trials each with zero expected return, the expected *maximum* Sharpe ratio grows unboundedly:

$$\mathbb{E}[\max_k S_k] \approx (1-\gamma) Z^{-1}\left(1-\frac{1}{K}\right) + \gamma Z^{-1}\left(1-\frac{1}{Ke}\right)$$

Implication: With enough trials, no Sharpe ratio threshold can reject a false strategy.

The Deflated Sharpe Ratio (DSR)

Corrects the observed SR for: sample length, return skewness & fat tails, and **number of strategy variations explored**.

Result: with **only 3 independent trials**, a 10-year backtest with $SR=1.0$ no longer reaches the 95% significance threshold.

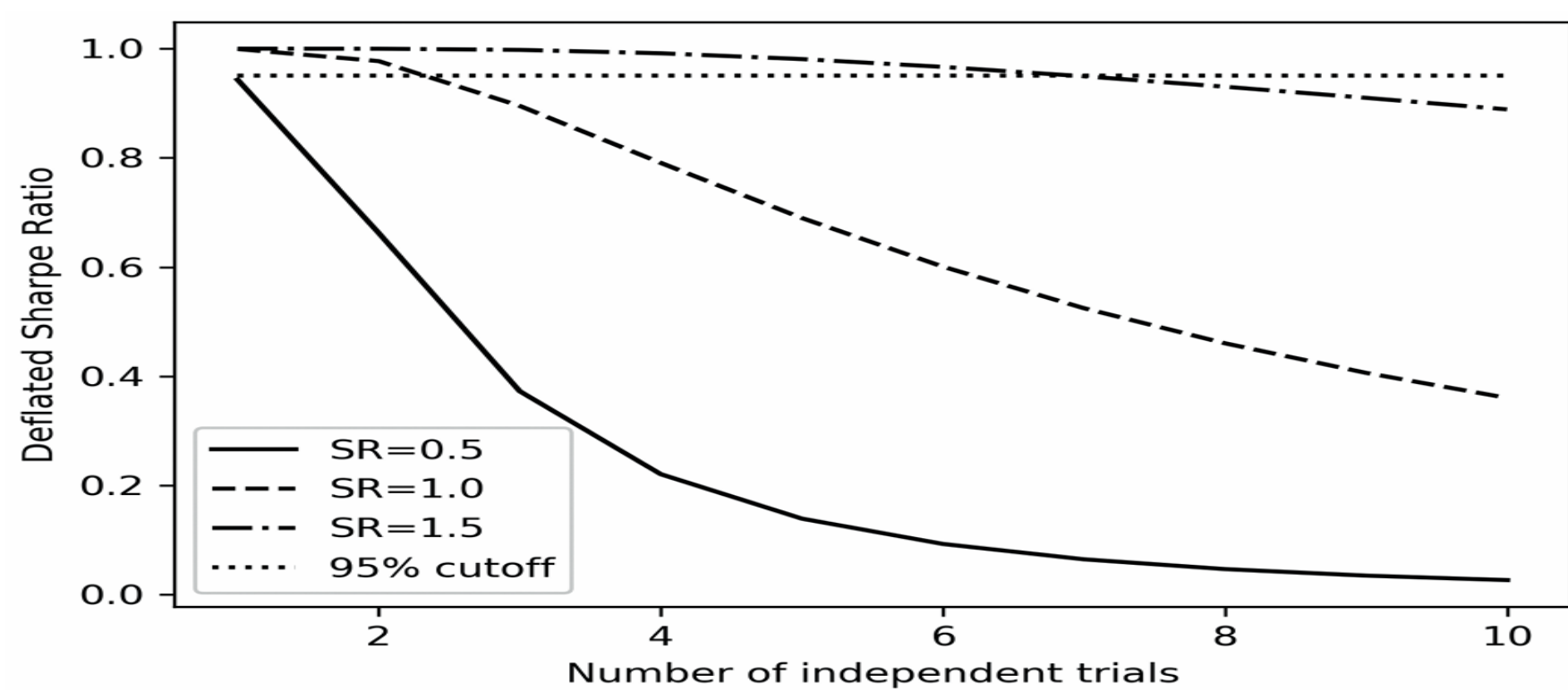
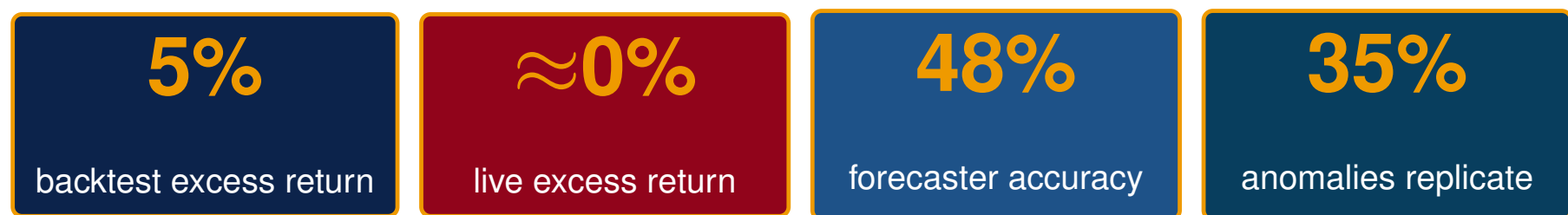


Figure 2: Deflated Sharpe ratios as a function of the number of trials, based on backtests of 10 years of independent and identically distributed normal daily returns.

Key Warning

Rule of thumb: The popular practice of “halving the backtest Sharpe ratio” has *no scientific basis*. The correct adjustment depends on the **number of trials run** – which is almost never disclosed in academic finance papers.

Results: Empirical Evidence



Sources: Brightman et al. (2015); Kaissar (2016); Hou, Xue & Zhang (2020).

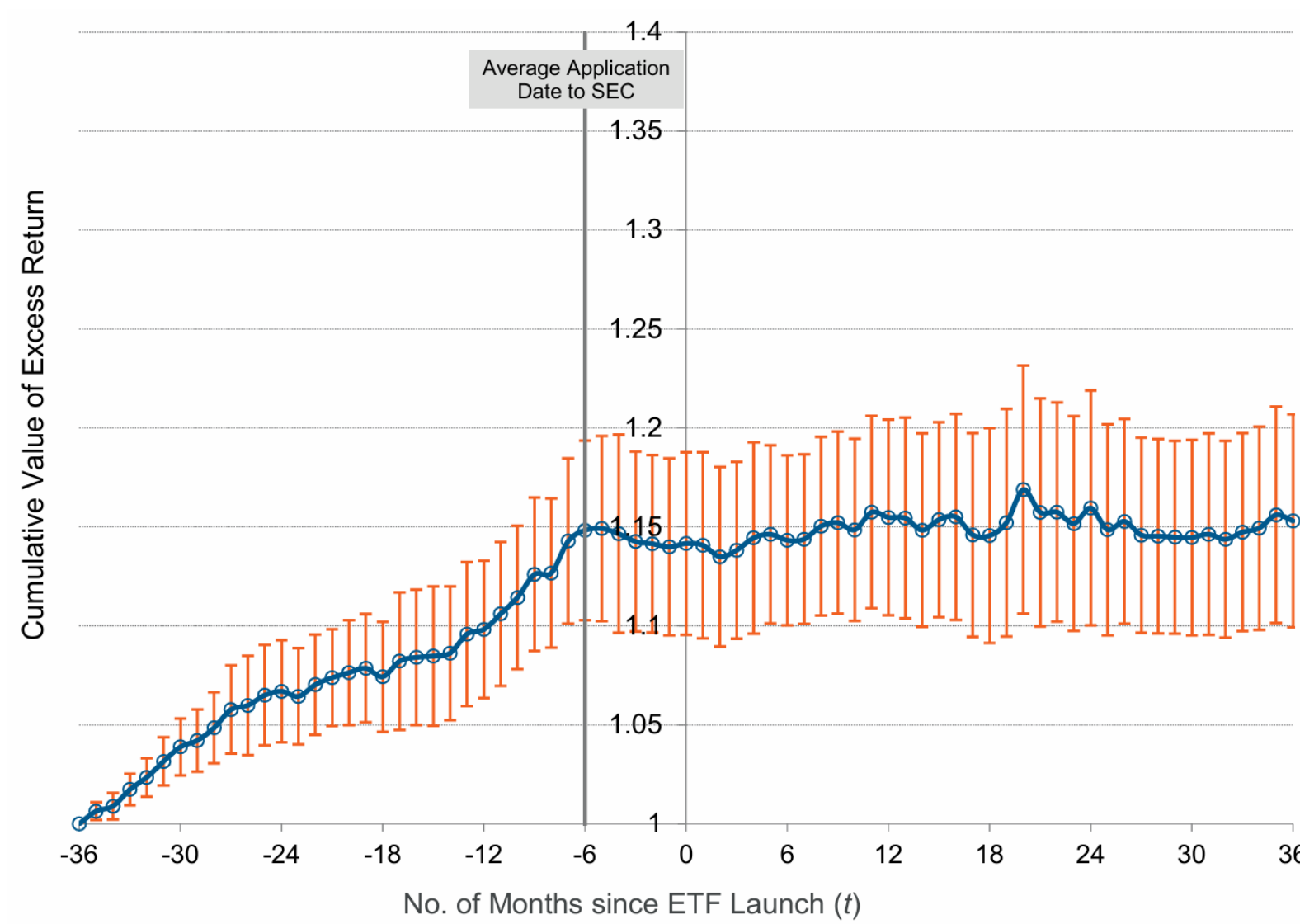


Figure 3: Backtested performance against performance out-of-sample

Anomalies & Forecasters

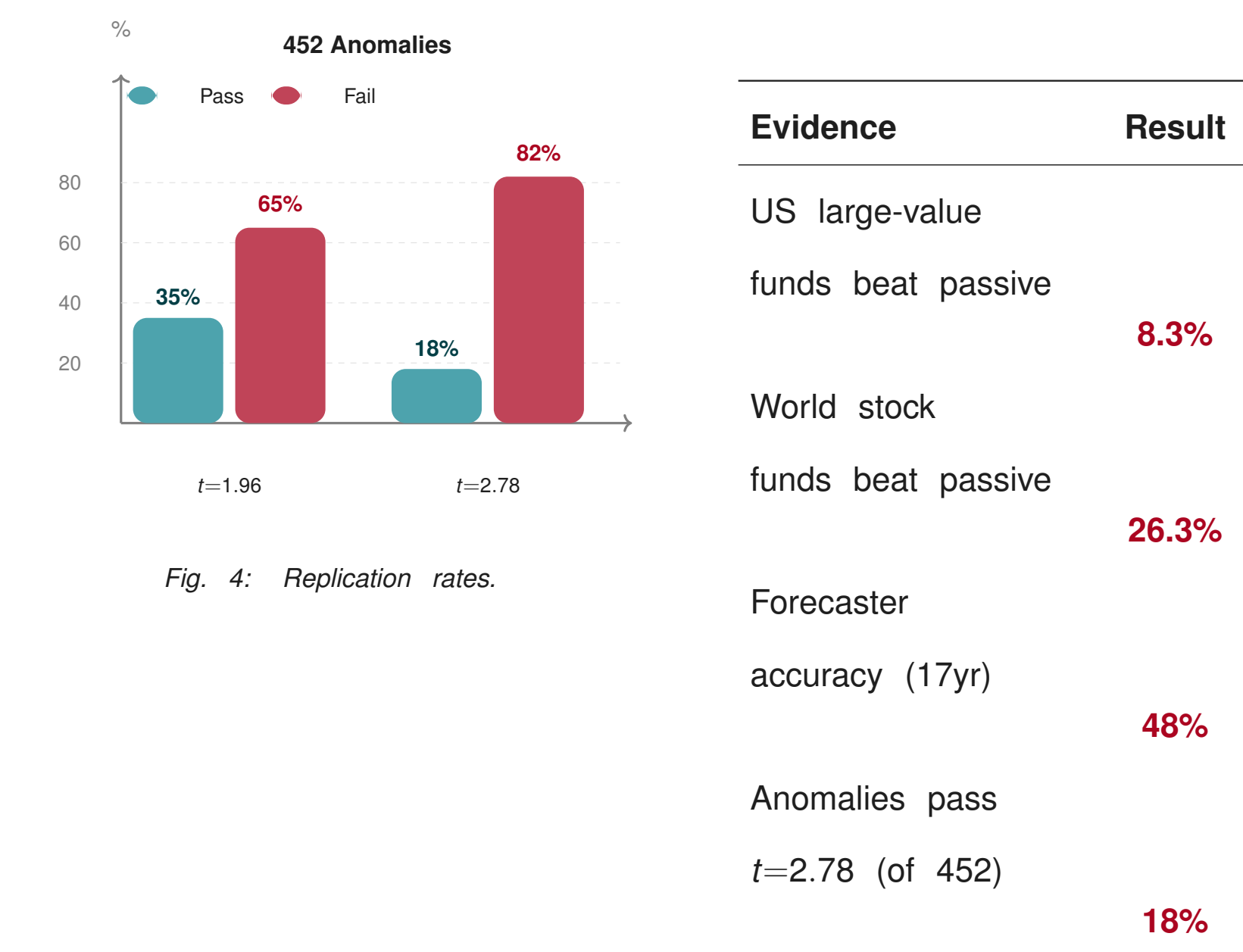
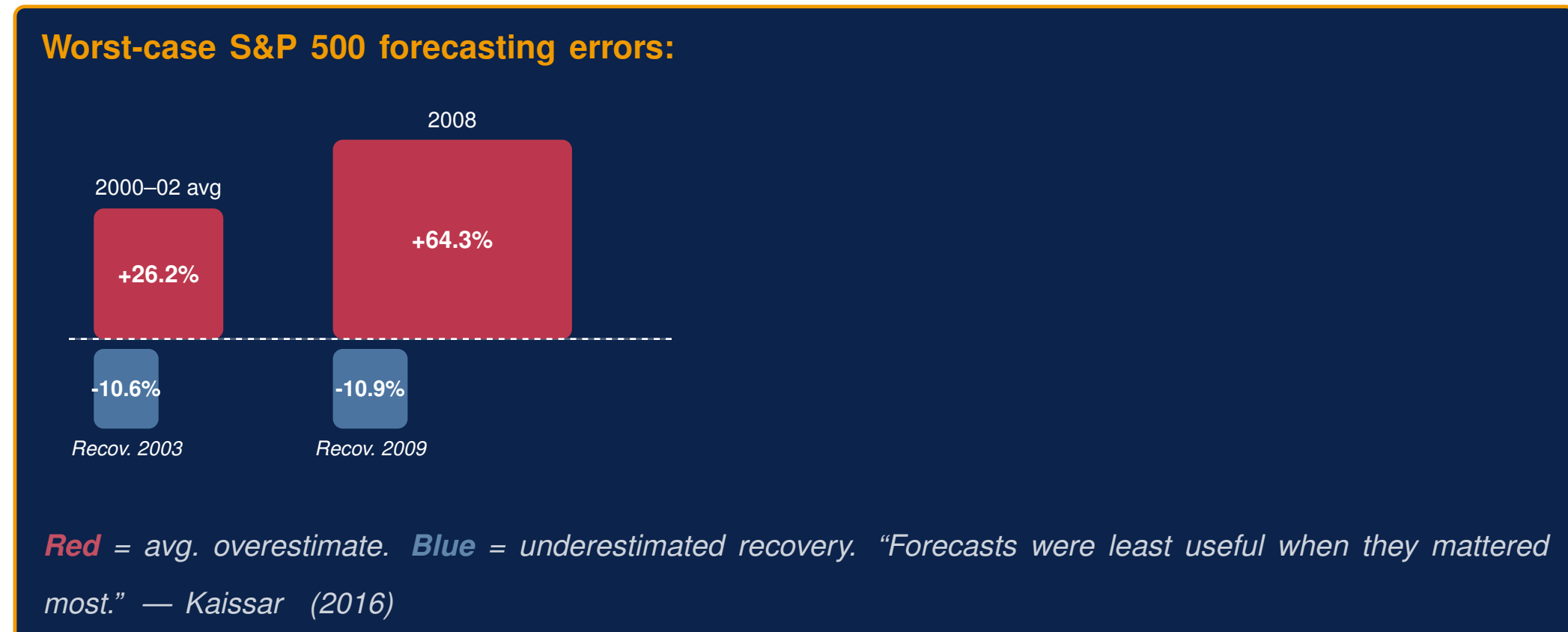


Table 1: All below expectation.



Discussion & Implications

Finance’s Replication Crisis

Backtest overfitting is the financial equivalent of **p-hacking**: running many analyses but reporting only the best outcome. Unlike biomedicine, finance lacks:

- Mandatory pre-registration of strategies
- Requirements to disclose number of trials run
- Controlled experiments to debunk false claims

Three Corrective Tools

1 Probabilistic Sharpe Ratio

Tests whether the observed SR is statistically significant given sample length – unlike the naive annualised SR.

2 False Strategy Theorem

Sets the hurdle any reported SR must clear, based on the *expected maximum* SR from K trials. Rejects false positives.

3 Deflated Sharpe Ratio (DSR)

Combines tools 1 & 2. Yields the probability an SR is genuinely positive, correcting for trials, skewness & kurtosis.

Limitations Acknowledged

- DSR assumes stationarity and ergodicity
- Some argue “factor investing” is less susceptible – authors disagree
- True discoveries may fade post-publication (McLean & Pontiff, 2015)

Conclusion

How widespread is backtest overfitting, and what tools can correct for it?

Finding 1 Overfitting is **pervasive**: ETFs, mutual funds, forecasters, and published anomalies all show evidence of selection bias under multiple testing.

Finding 2 PSR, False Strategy Theorem, and DSR can **detect and correct** for this bias, *if* the number of trials is disclosed.

Finding 3 “Halving the Sharpe ratio” has **no scientific basis**. With 3 trials, $SR=1.0$ over 10 years is already not significant.

“Trawling again and again through historical market data to identify an ‘optimal’ approach will often lead to a dead end.”

– Bailey & de Prado (2021)